

ECO362 Final 2024 Solutions

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Q1

(a)

Active management of your money can come with benefits, as the target-date funds use a fixed allocation between risky and safe assets, but “one-size” does not fit all.

$$w^* = \left(1 - \frac{H}{A}\right) \cdot \frac{\mathbb{E}_t [R_{m,t+1} - R_b | X_t]}{\gamma \cdot \sigma^2}$$

- The optimal weight on stocks depends on the human capital, which demonstrates large heterogeneity across individuals
- Investors have heterogeneity in risk-aversion, which also determines the optimal weight on stocks;
- Exploit expected return predictors, $\{X_t\}$ (like price-to-earning ratio or yields) to readjust the optimal weight on stocks;

Another accepted answer is that funds charge management fees (expense ratios)

(b)

Roth 401k is advantageous compared to the traditional if:

- tax rates are expected to increase in the future:
 - federally;
 - individually due to moving in the future to higher tax bracket or state with higher income taxes
- there are limits in contribution to the accounts; then, Roth allows you to put more of your savings in a tax sheltered account, as your contributed income is calculated after tax.
- due to risk-aversion agents prefer a fixed tax rate now than an uncertain one in the future

Grading:

- +3pt: each correct explanation for each subquestion

Q2

(a)

University i's excess return can be decomposed into:

$$R_{i,t} - R_{b,t} = \alpha_i + \beta_{i,t} \cdot (R_{m,t} - R_{b,t}) + \epsilon_{i,t}$$

Princeton's relatively low return can be driven by its:

- low or negative β_i , which captures low to negative correlation with systematic risk (proxied by the excess return of the benchmark)
- low idiosyncratic risk component, $\alpha_i + \epsilon_{i,t}$, where $\mathbb{E}[\epsilon_{i,t}] = 0$

Grading:

- +2 pt: correct identification of systematic risk and mapping to β
- +2 pt: correct identification of idiosyncratic risk and mapping to $\alpha + \epsilon$

(b)

The excess Princeton return is negative, $3.9\% - 5\% = -1.1\%$. Since $R_m - R_b = 10.1\% - 5\% = 5.1\% > 0$ and $\beta_i > 0$, a low systematic risk cannot be the only driver of Princeton's relatively low return. Negative α_i or $\epsilon_{i,t}$ are needed.

Grading:

- 4pt: correct answer by calculating the excess return of benchmark to be positive and excess return of Princeton to be negative
- 1pt: correct without calculating excess return, but explicitly mentioning that there needs to be a negative idiosyncratic risk component

(c)

No, because skill is measured using the information ratio, $\frac{\alpha_i}{\sigma_i(\epsilon_i)}$, which cannot be determined with the given information.

Grading:

- 4pt: correct answer that skill, which can be measured by the information ratio, cannot be determined

(d)

Under the assumption of equal betas and assuming that the funds share the same idiosyncratic variance, we can estimate it by calculating the sample variance of the returns,

$$\text{Var}(\epsilon_{i,t}) = \text{Var}(\alpha_i + \epsilon_{i,t}) = \text{Var}(R_i)$$

Hence,

$$\sigma(\hat{\epsilon}_i) = \sqrt{\tilde{X}^2 - (\bar{X})^2} \simeq 2.5\%$$

where $\bar{X}$ is the sample average of returns across universities.

Grading:

- 4pt: correct calculation
- 1pt: relating the idiosyncratic standard deviation to the standard deviation of the returns, but making calculation error

Q3

(a)

The annually compounded return formula is

$$r = \frac{P_t + X_t}{P_{t-1}} - 1 = \frac{294.11 + 0.16}{130.55} - 1 = 2.25 - 1$$

So the return is 125%.

Grading:

- 4pt: Correct
- 2pt: Up to one error in return formula (didn't subtract 1, didn't take into account dividends, etc.)
- 0pt: Incorrect/blank

(b)

The cumulative return formula over multiple periods with dividends is

$$r = \prod_{t=2020}^{2023} \frac{P_t + X_t}{P_{t-1}} - 1 = \frac{294.11 + 0.16}{130.55} \times \frac{146.14 + 0.16}{294.11} \times \frac{495.22 + 0.16}{146.14} - 1 = 3.80 - 1 = 2.80$$

So assuming dividends are reinvested in stock purchases, the cumulative return is 280%.

Grading:

- 4pt: Correct
- 2pt: Attempted to calculate cumulative returns, but had errors in calculations (didn't take into account dividends correctly, didn't subtract 1, didn't cumulate returns correctly, etc.)
- 1pt: Calculated cumulative return as if it were annual
- 0pt: Incorrect/blank

(c)

Using the CAPM formula:

$$E(R_i) = R_f + \beta_i[E(R_m) - R_f] = 5\% + 2(9\% - 5\%) = 5\% + 2(4\%) = 13\%$$

So the expected annual return is 13%.

Grading:

- 4pt: correct
- 3pt: correct equation, small calculation error
- 2pt: almost correct equation but forgets some component (like adding/subtracting the risk-free rate in the right places)
- 1pt: tried to specify but some errors in the equation and answer
- 0pt: no coherent answer

(d)

Using the Gordon growth model:

$$\frac{P_i}{D_i} = \frac{1 + G_i}{\bar{R}_i - G_i} \implies G_i = \frac{(P_i/D_i) \bar{R}_i - 1}{1 + (P_i/D_i)} = \frac{\frac{495.22}{0.16} 13\% - 1}{1 + \frac{495.22}{0.16}} = 12.9635\%$$

So the expected annual dividend growth is approximately 12.96%.

Grading:

- 4pt: correct
- 3pt: correct equation and algebraic manipulation, but small calculation error
- 2pt: correct equation, but algebra errors OR major substitution errors like using the wrong date
- 1pt: tried to specify but some errors in the equation and answer
- 0pt: no coherent answer

(e)

With $g = 12.86\%$ (reduced by 0.1%), the new price is:

$$P = \frac{D(1 + G^{new})}{\bar{R} - G^{new}} = \frac{0.16(1.1286)}{0.13 - 0.1286} = \frac{0.18058}{0.0014} = 128.98$$

So the new stock price would be \$128.98. We conclude that Nvidia's stock price is extremely sensitive to the dividend growth rate, which should track the expected growth of the AI industry.

Grading:

- +2pts for correct price (partial credit given for small mistakes)
- +1pt for correct answer (high sensitivity)
- +1pt for good explanation (connecting expected dividend growth with expected growth of the AI industry)
- 0pt: no coherent answer

Q4

(a)

Using the Gordon growth model, the value of human capital H is

$$\frac{H}{L_0} = \frac{1 + G}{R - G} = \frac{1 + 0.02}{0.05 - 0.02} = 34$$

$$H = 34 * \$50,000 = \$1,700,000$$

Grading:

- 4pt: correct answer and equation
- 2pt: correct use of the Gordon growth model and calculation of H/L_0 (right parameters) but wrong calculation for H
- 1pt: correct equation for the Gordon growth model but wrong parameter (e.g., used the mean of stock return instead of the riskless rate for R).
- 0pt: incorrect (including the case of the incorrect equation and calculation but correct answer)

(b)

The optimal face amount of life insurance of the mean-variance expected utility maximizer is the amount of human capital ($F = H$). Thus, $F = \$1,700,000$. Regardless of your financial wealth or risk aversion, the variance is minimized when $F = H$. Thus, it does not depend on financial wealth or risk aversion. Answers that equalized F to $H \cdot (1 + R_b)$ were also taken as correct.

Grading:

- +2pt: correct amount of F (no penalty for the carry-over mistake for H if the same H calculated in 4(a) was used)
- +2pt: correct answer for the second part (whether it depends on financial wealth or risk aversion)
- +1pt: Partially correct answer for the second part (either correct about financial wealth or risk aversion)

(c)

If we focus on the fact that this calculation is an approximation since we ignore retirement, it is more accurate for younger people since they have more years left until their retirement. (In this model, we assume $t \rightarrow \infty$. It is a better approximation if the actual t is large, e.g., 30 years, compared to $t = 5$.)

Grading:

- 4pt: correct answer and reasoning
- 2pt: correct answer but inadequate reasoning (not explaining why it can be the bad approximation for older people since we ignore retirement in Gordon growth model)
- 0pt: incorrect
- Note: the key point in this problem was whether you can acknowledge the limitation of applying the Gordon growth model to calculating human capital since it assumes that people will not retire at all. Some common incorrect answers: more accurate for the older person since he/she is more likely to die soon/ have more dependents/ easy to estimate incomes (not considering the effect of retirement)

(d)

The optimal allocation to stocks as a share of financial wealth is

$$\frac{S}{A} = \left(1 + \frac{H}{A}\right) \frac{\mu - R_b}{\gamma \sigma^2} = \left(1 + \frac{1,700,000}{1,000,000}\right) \frac{0.09 - 0.05}{10 * 0.2^2} = 0.27$$

Grading:

- 4pt: correct answer and equation (no penalty for the carry-over mistake for H if the same H calculated in 4(a) was used)
- 2pt: correct equation but calculation error or wrong parameters
- 1pt: Correct equation of (S/A) but wrong parameters and calculation error
- 1pt: Using the optimal stock weight equation without considering human capital ($\frac{\mu - R_b}{\gamma \sigma^2}$). 0 pt when the calculation is wrong in this case.
- 0pt: Incorrect

Q5

(a)

The term structure model is:

$$y(t) = \bar{y} + \frac{1}{t} \left(\frac{1 - \rho^t}{1 - \rho} \right) (y(1) - \bar{y}) + \frac{t-1}{t} \mu$$

Solving for $y(2)$:

$$y(2) = 5\% + \frac{1}{2} \left(\frac{1 - 0.8^2}{1 - 0.8} \right) (5\% - 5\%) + \frac{2 - 1}{2} 0.5\% = 5.25\%$$

Solving for $y(3)$:

$$y(3) = 5\% + \frac{1}{3} \left(\frac{1 - 0.8^3}{1 - 0.8} \right) (5\% - 5\%) + \frac{3 - 1}{3} 0.5\% = 5.33\%$$

Grading:

- +2 pt: Correct calculation for $y(2)$
- +2 pt: Correct calculation for $y(3)$

(b)

By the expectations hypothesis, $\mathbb{E}[r_1(t)] = \mu + y_0(1) = 5.5\%$, where μ and $y_0(1)$ are constants. This means both of the zero coupon bonds from (a) offer the same expected return. Note that although they offer the same expected return, the 3-year bond has a higher yield, so that is compensated for by a lower capital gain than the 2-year bond.

Grading:

- 4pt: Uses the expectations hypothesis correctly to conclude they have same expected returns
- 1pt: Partial credit for explaining that expected return is comprised of yield and capital gain/loss term
- 0pt: Incorrect (note: yield is not expected return, so answers directly relating yield to expected return are incorrect)/Provides conflicting answers

(c)

The term structure model slope depends on two effects:

- Expectation of future short rates:
 - In this case, because $y_0(1) = \bar{y}$, this effect is 0
- Term premium
 - The term premium is positive and makes the yield curve more upward sloping. As t increases, the term premium grows closer to μ .

Overall, the term structure is upward sloping.

Grading:

- +2 pt: For explaining how each effect of the term structure model impacts the slope
- 1 pt: Identifies upward sloping term structure because the yields are observed to be increasing, but does not provide additional explanation concerning why

(d)

The fixed rate is the par yield on a 3-year par bond, which can be calculated using the par yield formula. We assume a face value of \$1.

$$c(3) = \frac{1 - P_b(3)}{P_b(1) + P_b(2) + P_b(3)} = 0.0547$$

We calculate zero-coupon bond prices:

$$P_b(1) = \frac{1}{e^{0.05 \cdot 1}} = 0.95123$$

$$P_b(2) = \frac{1}{e^{0.0525 \cdot 2}} = 0.90003$$

$$P_b(3) = \frac{1}{e^{0.0533 \cdot 3}} = 0.85222$$

The par-yield is 5.47%.

Grading:

- 4 pt: Correct
- 2 pt: Incorrect answer but otherwise correct approach
- 2 pt: Correct approach except for using annual compounding
- 1 pt: Errors in calculations of par yield, but wrote the formula and tried to compute ZCB prices
- 0 pt: Incorrect

(e)

The futures price under no arbitrage is:

$$F = S_0 e^{r_b \cdot T} = S_0 e^{0.05 \cdot 1} = 57.70$$

where S_0 is the current price of the 10-year zero coupon bond that will be exchanged a year from now, which is equivalent to the price of an 11-year zero coupon bond:

$$S_0 = \frac{100}{e^{y_b(11) \cdot 11}} = 54.88$$

We calculate $y_b(11)$ using the term structure model to obtain $y_b(11) = 5.45\%$. The futures price is \$57.70.

Grading:

- 4 pt: Correct
- 3 pt: Correct approach but used 10-year instead of 11-year yield

- 2 pt: Up to 2 errors: used annual compounding, 10-year instead of 11-year yield, calculator error, wrong time index, etc.
- 0 pt: Incorrect

Q6

(a)

Recall the call-put parity equation, $C = P + S_0 - Xe^{-r_b T}$ (we derived this by replicating the call option by buying a put option with 4-month maturity ($T = \frac{1}{3}$), and strike price $X = \$145$, buying the stock at time 0 for S_0 , and borrowing $Xe^{-r_b T}$).

$$\begin{aligned} P(T = \frac{1}{3}, X = \$150) &= C(T = \frac{1}{3}, X = \$150) - (S_0 - Xe^{-r_b T}) \\ &= \$5.43 - \$150(1 - e^{-0.05/3}) = \$5.43 - \$2.4793 \approx \$2.95 \end{aligned}$$

The price of a put option on the stock with 4-month maturity and strike price of \$150 is \$ 2.95 .

Grading:

- 4pt: correct
- 3pt: correct equation (call-put parity) but small calculation error
- 2pt: correct equation but missing/incorrect calculation
- 1pt: tried to specify but some errors in the equation and answer
- 0pt: no coherent answer

(b)

We can again use put-call parity to get an answer:

Since the payoffs on a long-position on an option are non-negative, we must have that $P \geq 0$. This gives us a lower bound on the price of the call option:

$$\begin{aligned} C(T = \frac{1}{3}, X = \$145) &= P(T = \frac{1}{3}, X = \$145) + S_0 - Xe^{-r_b T} \\ &\geq S_0 - Xe^{-r_b T} = \$150 - \$145e^{-0.05/3} = \$150 - \$142.6034 \approx \$7.40 \end{aligned}$$

Based on the information given, the tightest lower bound on the price of the call option with 4-month maturity and strike price of \$145 is \$7.40

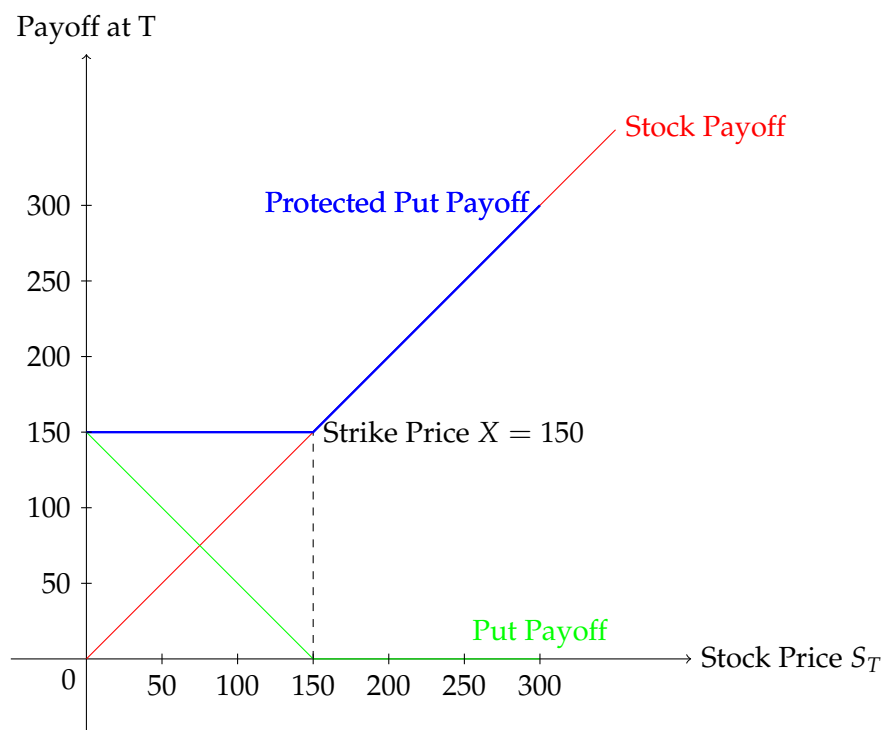
Note that there are other lower bounds we can find, but they are less tight than this. For example, we can note that the payoff of the missing call option is given by $\max(S_{\frac{1}{3}} - 145, 0)$. Note further that this is larger or equal to $\max(S_{\frac{1}{3}} - 150, 0)$ in any state of the world. Hence, we must have at least that $C(\frac{1}{3}, 145) \geq C(\frac{1}{3}, 150) = \5.43

In this class, we have only talked about European options, which can only be exercised at maturity. American options are those which investors can choose to exercise before maturity, and they generally trade at a higher price due to additional flexibility they provide. Answers that pointed to $C(T = \frac{1}{6}, X = \$145) = \6.98 as a lower bound (since a 4-month American option could be exercised after 2 months) were given partial credit only when the reasoning was thoroughly explained.

Grading:

- 4pt: correct
- 3pt: correct equation (call-put parity with $P \geq 0$) but small calculation error
- 2pt: answers lower bound that is not tight (\$ 5.43 or \$6.98), explained well OR correct equation but missing calculation
- 1pt: tried to specify but some errors in the equation and answer
- 0pt: no coherent answer

(c)



You can find these diagrams on slide 9 of the Options lecture.

Grading:

- +1pt: both axes labelled correctly
- +1pt: correct overall shape
- +1pt: payoff = X for $S_T < X$

- +1pt: payoff = S_T for $S_T < X$
- Multiple points may be taken off if the student draws multiple graphs and does not label them appropriately, even if one of the graphs might be correct.

(d)

This strategy is called a straddle.

Suppose we are buying both a put and a call option for the same maturity and strike price. This means that if the stock price is above the strike price, we exercise the call option and get $S_T - X$, and if the stock price is below the strike price, we exercise the put option and sell, receiving $X - S_T$ as our payoff.

Note that the payoff can be written as

$$\max(S_T - X, 0) + \max(X - S_T, 0) = |S_T - X|$$

i.e. we get the stock's absolute deviation from X as our payoff. If we buy a straddle, we are betting that the volatility on the asset is going to be high and that the stock price will deviate from the strike price significantly, at least in expectation. If you believe that volatility is going to be low, you would take a short position on the straddle. See Options slides 10-12.

Grading:

- +1pt: identifying the strategy as a straddle
- +1pt: describe straddle payoff at T (equation or case-by-case description are both fine as long as they are complete)
- +1pt: mention "bet on high volatility" or equivalent
- +1pt: explanation of "bet on high volatility"